PROJECT & TEAM · HEDERA CROSS CAMPUS CHALLENGE 2026

EL JOKER

5ALIHA 3LINA.

The control layer for AI agents that spend money. One AI proposes, four strict agents verify, decide and pay. Above the limit, only a human co-signature on Hedera opens the vault.

TEAM
CHAIN619

MEMBERS
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PROBLEM

THREE WAYS AGENTS FAIL

01



HALLUCINATE

Invents a supplier, a price or an approval, and acts on it.

02



OVERSPEND

Optimizes the task, ignores the budget, or pays the same invoice twice.

03



HIDE THE TRAIL

Acts first, explains later. Its logs can be edited after the fact.

Today you choose: autonomy or control. We remove that trade-off, for any organization that lets an agent spend.

SOLUTION

AI PROPOSES. RULES DECIDE. HEDERA PROVES.

Autonomy for the routine. Hard limits on the money. Proof for everyone.



PLAN

Lem3alem · Coordinator

Turns the request into a clear mission.

PROPOSE

Lemhaf · AI

Claude ranks the supplier offers.

VERIFY

Nasnes

Checks registry, evidence and the AI's claims.

DECIDE

3ases · no LLM

Fixed rules per organization: approve, block or escalate.

CONTROL

A human

Above the limit, co-signs on Hedera to release the money.

PAY & PROVE

Kanawita · Hedera

Pays only if authorized. Every step on HCS.

PRODUCT / POC · HOW A MISSION FLOWS

REQUEST → PROOF

Every handoff is visible. Every decision leaves a receipt on Hedera.

REQUEST	PLAN	RESEARCH	VERIFY	GUARD	PAY	PROOF
The operator types the need	**Lem3alem** Valid mission, no guessed facts	**Lemhaf** Claude ranks the offers	**Nasnes** 10 independent checks	**3ases** Approve, block or human	**Kanawita** On testnet, exactly once	**Hedera** Every step on HCS

HUMAN CO-SIGNATURE

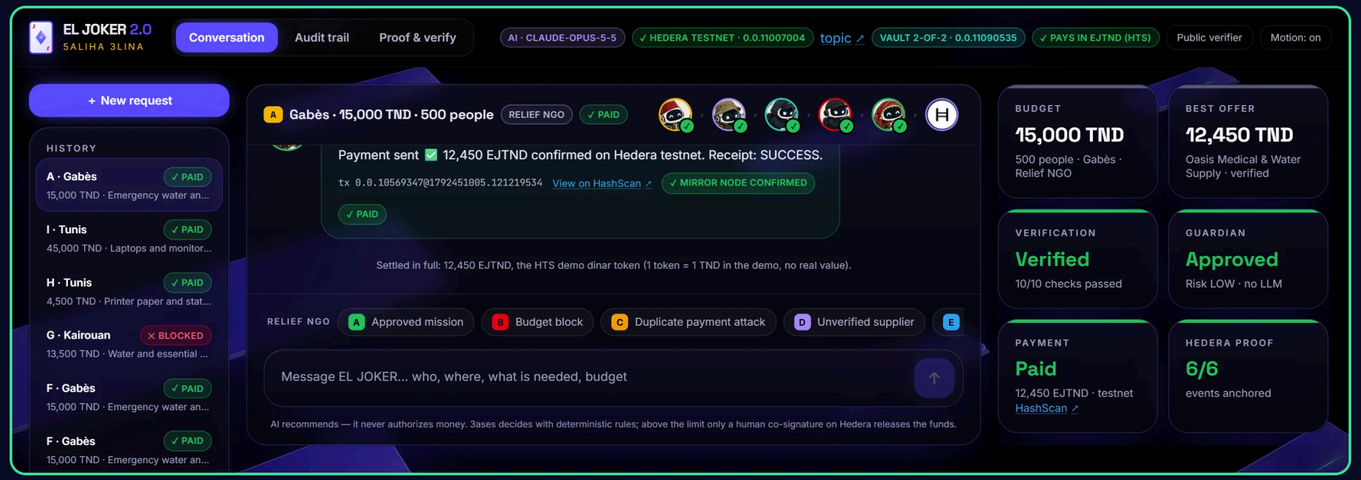
Above the limit: locked in a 2-of-2 Hedera vault until a human signs.

PRODUCT / POC · LIVE DEMO

15,000 TND BUDGET. MISSION CLEARED.

APPROVED

- Claude ranks 3 offers: **Oasis, 12,450 TND**
- Nasnes 10/10 checks · 3ases 12/12 checks, risk LOW
- Kanawita pays **12,450 EJTND**, our test dinar on Hedera
- 6 decisions anchored on Hedera, confirmed by the Mirror Node



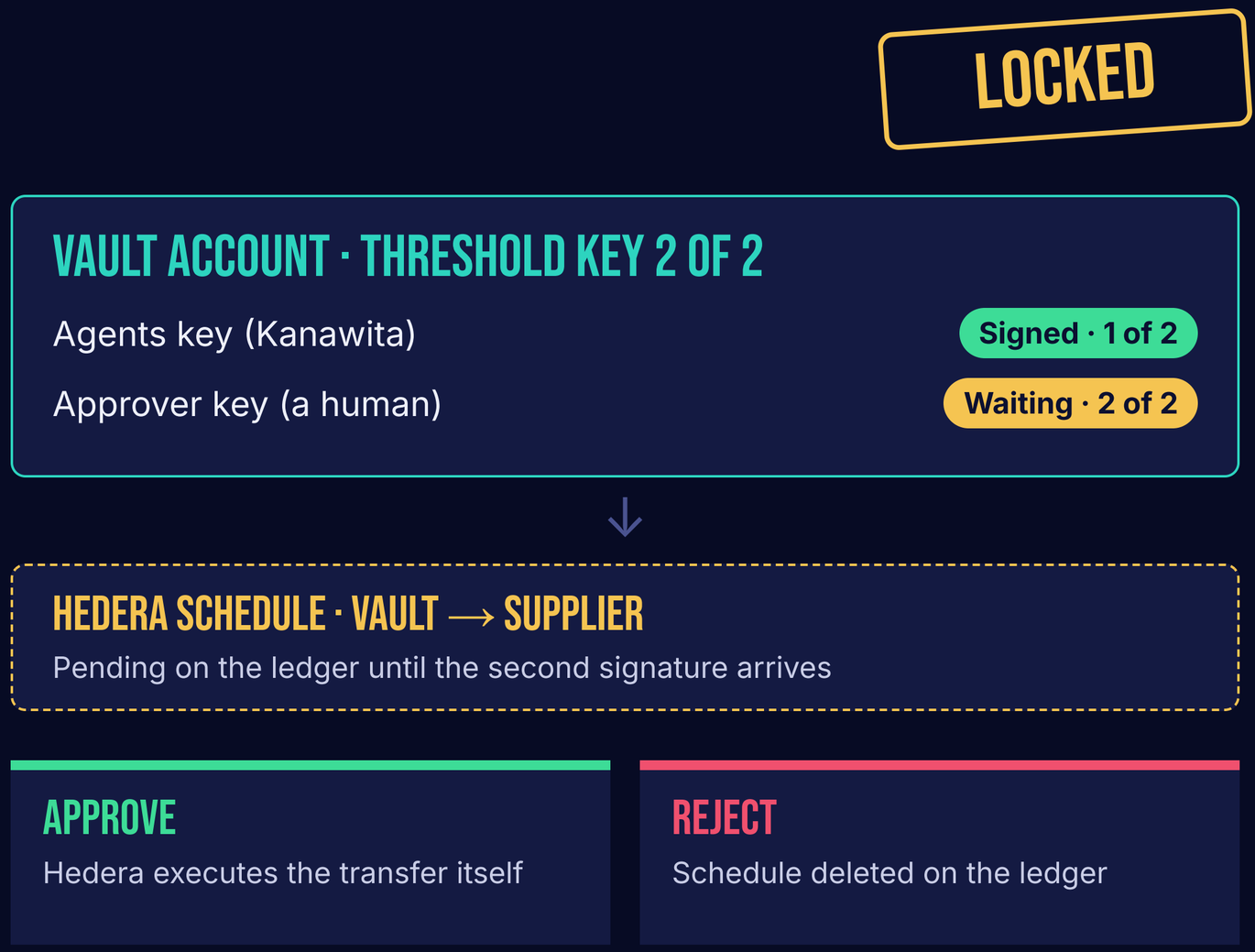
Live run on Hedera testnet with Claude Opus 5.5: 12,450 EJTND paid, 6/6 proofs anchored

PRODUCT / POC · HUMAN CO-SIGNATURE

ABOVE THE LIMIT, HEDERA HOLDS THE MONEY.

- ACME: **35,460 TND** of laptops, over its 6,000 TND limit
- Kanawita can only **lock** it: a Hedera schedule, 1 of 2 signatures
- Operator + reason + PIN → the approver signs → Hedera pays
- Reject → the schedule is deleted. Nothing moves

The agents' key alone can never move the vault. The approval is bound to the exact facts reviewed.



TECHNOLOGY & HEDERA

FOUR HEDERA SERVICES



No smart contract: Hedera enforces the human signature, carries the money and lets anyone check.

CONSENSUS SERVICE

Every decision fingerprinted (SHA-256), chained to the last one and published. Topic 0.0.11007004.

SCHEDULED TX + KEYS

The 2-of-2 vault: above the limit, the human co-signature is enforced by the ledger itself.

TOKEN SERVICE

EJTND, our test dinar: the full amount is paid in dinars, fixed supply, no admin key.

MIRROR NODE

Anyone re-checks everything, even from a browser, without trusting our server.

ARCHITECTURE

WEB APP
Next.js, in the browser

→

API · 5 AGENTS
Express + SQLite, Claude for Lemhaf only

→

HEDERA
HCS · schedules · HTS, via the SDK

→

MIRROR NODE
The history, free to read

→

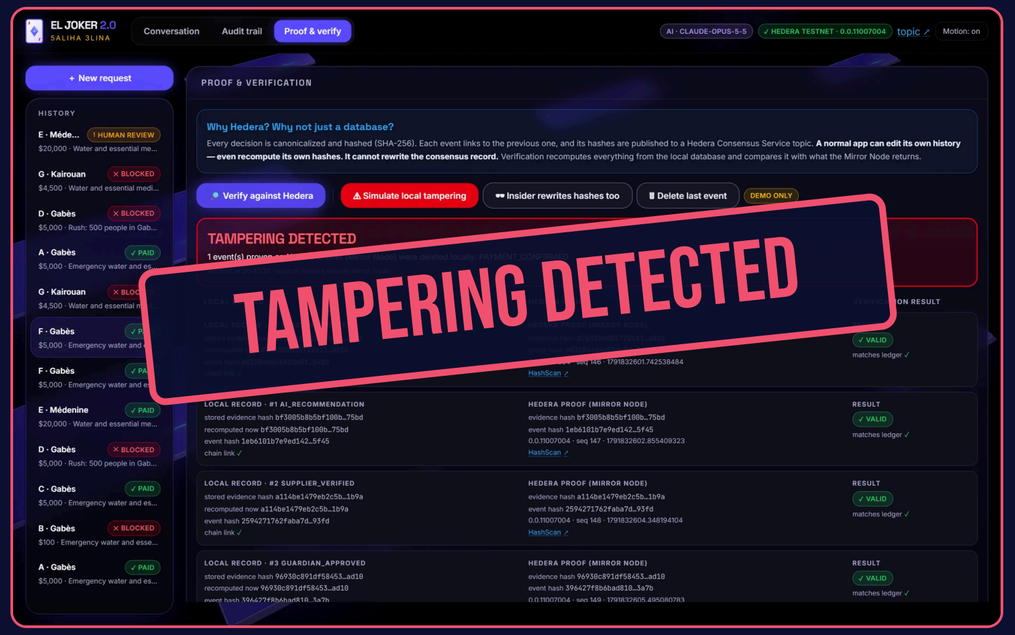
PUBLIC VERIFIER
Any browser re-checks a mission

TECHNOLOGY & HEDERA · WHY NOT A DATABASE

A DATABASE CAN REWRITE ITS PAST. HEDERA CAN'T.

Insider attack (demo)	Local check	With Hedera
Edit the approved price	Caught	Caught
Edit it and recompute every hash	Missed	Caught
Delete the payment record	Missed	Caught

Verified live on testnet. And you don't have to trust us: our **public verifier** recomputes every hash in your browser, reading Hedera directly.



Local record vs Hedera proof vs verification result

IMPACT & ADOPTION

WORKS TODAY. WHO NEEDS IT.



9/9
scenarios passed live on Hedera testnet

115
automated tests, incl. 7 in a real browser

4
Hedera services: HCS, schedules, HTS, Mirror Node

2
organizations on one engine: an NGO and a company

FIRST
NGOS & GRANTS
Must justify every dinar to their donors

THEN
COMPANIES
Agents in procurement, within the spending policy

FOR EVERYONE
DONORS & AUDITORS
Verify any mission themselves, on Hedera

PLUG IN
AGENT PLATFORMS
Add rules, a human signature and proof to any agent

Who pays: any organization that lets agents spend — SaaS per organization + per proof, NGOs first, then companies (our hypothesis).
Honest limits today: fictional suppliers, a test dinar token, a server-held approver key, a demo PIN.

VISION

EVERY AI AGENT THAT SPENDS MONEY, UNDER CONTROL.

NOW

PILOT-READY

Real suppliers, a licensed partner for the dinar token

NEXT

KEYS IN WALLETS

Approvers sign from their own wallet, real identity

THEN

TWO PILOTS IN TUNISIA

One NGO and one company, with real money flows

SCALE

POLICY LIBRARY

Ready-made policies per sector, shared supplier registry

The trust layer for agentic payments on Hedera. Starting in Tunisia.



CONCLUSION

AUTONOMOUS, BUT NEVER UNCONTROLLED.

WHY IT MATTERS

AI agents now spend real money. Organizations need control and proof.

WHAT IS DIFFERENT

The AI never decides. Hedera enforces the human signature and proves every step.

WHAT IS NEXT

Two pilots in Tunisia: one NGO, one company. Then any agent, any sector.

EL JOKER — 5ALIHA 3LINA

THANK YOU

github.com/rihamghoudi-bit/el-joker-2 · hashscan.io/testnet/topic/0.0.11007004 · verify any mission: /verify